

Probability Distributions in R

In this fourteenth article in the R series, we shall explore the common probability distribution functions available in R.



We will use version 4.2.1 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version
```

```
R version 4.2.1 (2022-06-23) -- "Funny-Looking Kid"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

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Consider the *mtcars* data set in the lattice library:

```
> head(mtcars)
      mpg cyl disp  hp drat   wt  qsec vs am gear carb
Mazda RX4   21.0   6  160 110 3.90 2.620 16.46  0   1    4    4
Mazda RX4 Wag 21.0   6  160 110 3.90 2.875 17.02  0   1    4    4
Datsun 710  22.8   4  108  93 3.85 2.320 18.61  1   1    4    1
Hornet 4 Drive 21.4   6  258 110 3.08 3.215 19.44  1   0    3    1
Hornet Sportabout 18.7  8  360 175 3.15 3.440 17.02  0   0    3    2
Valiant    18.1   6  225 105 2.76 3.460 20.22  1   0    3    1
```

The *summary()* function can be used to produce the basic statistics of the data, as shown below:

```
> summary(mtcars)
      mpg          cyl          disp          hp
Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0
1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5
Median :19.20   Median :6.000   Median :196.3   Median :123.0
```

```
Mean      :20.09   Mean      :6.188   Mean      :230.7   Mean      :146.7
3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0
Max.      :33.90   Max.      :8.000   Max.      :472.0   Max.      :335.0
```

The Tukey's five number summary (minimum, lower-hinge, median, upper-hinge, and maximum) can be obtained using the *fivenum()* function, as follows:

```
> mtcars$disp
[1] 160.0 160.0 108.0 258.0 360.0 225.0 360.0 146.7 140.8
167.6 167.6 275.8
[13] 275.8 275.8 472.0 460.0 440.0 78.7 75.7 71.1 120.1
318.0 304.0 350.0
[25] 400.0 79.0 120.3 95.1 351.0 145.0 301.0 121.0

> fivenum(mtcars$disp)
[1] 71.10 120.65 196.30 334.00 472.00
```

A histogram of the *mtcars* displacement data is shown in Figure 1.

The *stem()* function can generate a stem-and-leaf plot for input data. For example, we can obtain a plot for the *mtcars* cylinder, as demonstrated below:

```
> mtcars$cyl
[1] 6 6 4 6 8 6 8 4 4 6 6 8 8 8 8 8 4 4 4 4 8 8 8 8 4 4 4
8 6 8 4
```

```
> stem(mtcars$cyl)
```

The decimal point is at the |

```
4 | 000000000000
4 |
5 |
5 |
6 | 00000000
6 |
7 |
7 |
8 | 0000000000000000
```

Uniform

The Uniform distribution has the following density function: